Adversarial Example Generation

Qualification Task Report

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Task 1: Implementation

A simple fully-connected neural network was trained on MNIST — 60 000 training images of handwritten digits — for 10 epochs using Adam and cross-entropy loss. As shown in Figure 1, loss dropped from 0.40 to 0.025 while training accuracy climbed from 88% to 99.1%. The biggest jump happened in the first two epochs, after which progress slowed down noticeably. On the held-out test set the model reached **97.7%** accuracy — a healthy result with no signs of overfitting.

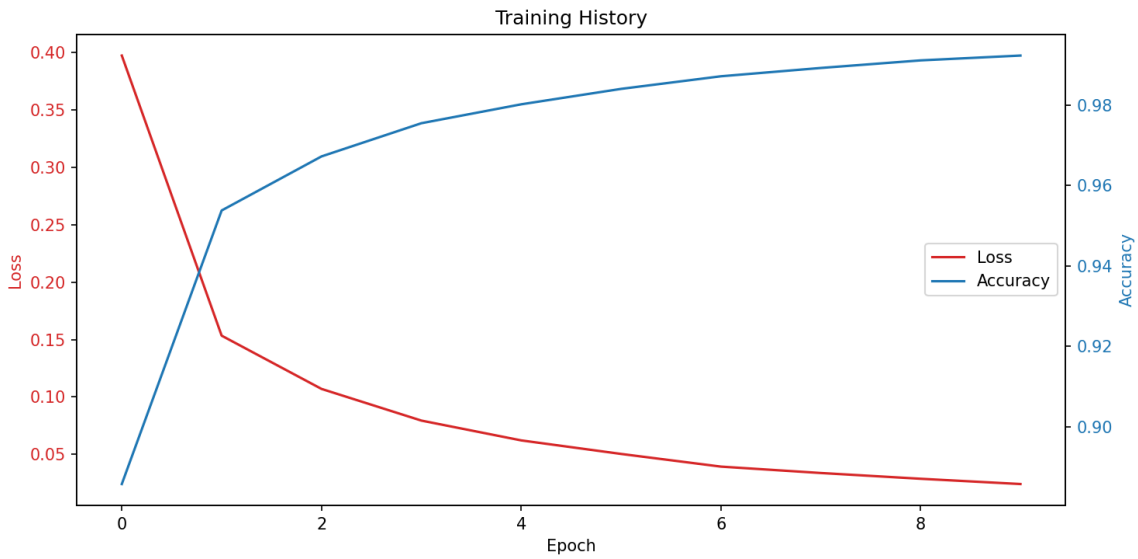

Figure 1: Training loss and accuracy over 10 epochs.

Next, the FGSM attack was implemented. Instead of adjusting model weights to minimise loss, FGSM nudges the input image to *maximise* it:

$$x' = x + \varepsilon \cdot \text{sign}(\nabla_x J(\theta, x, y))$$

With $\varepsilon = 0.1$ on 50 test images, accuracy collapsed to **2%**. The digits still look perfectly readable to a human — the model is just completely fooled (Figure 3). One small bug

along the way: the first visualisation attempt produced a noisy strip (Figure 2) because the flat [50×784] tensors were passed to `make_grid` without reshaping back to [50×1×28×28]. Fixed with a single `.view()` call.

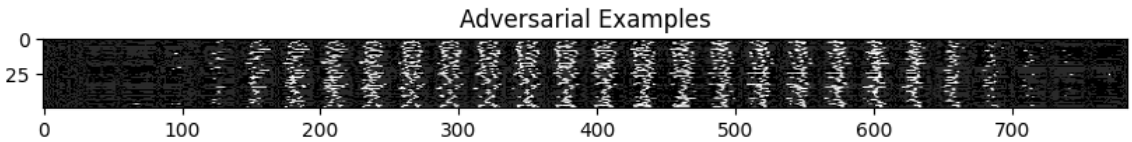

Figure 2: Broken visualisation before the reshape fix.

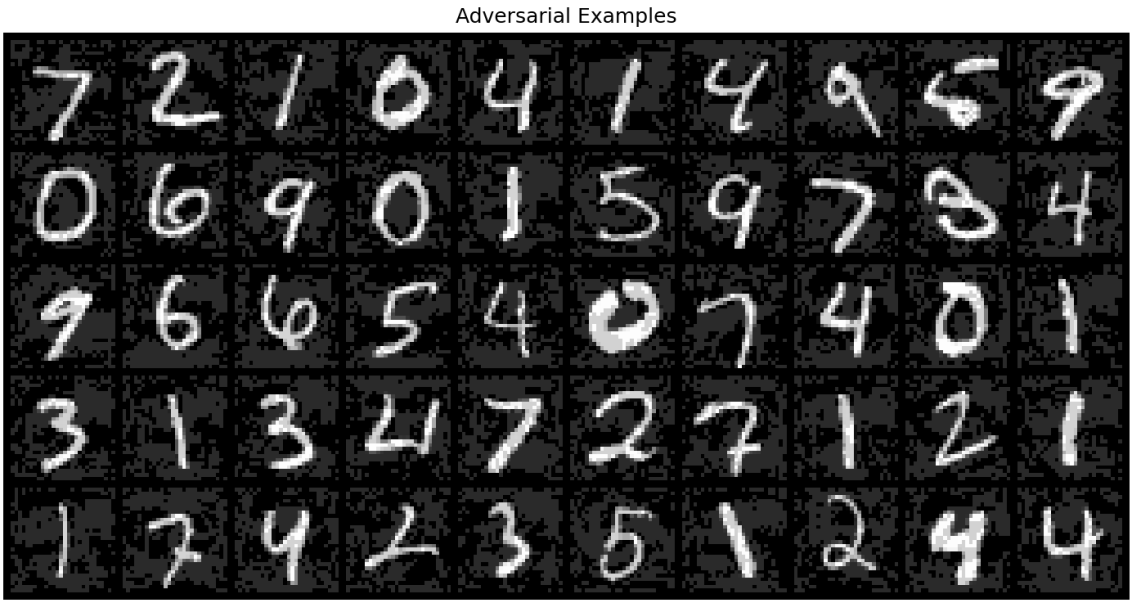

Figure 3: Adversarial examples after the fix — digits are readable to humans, but the model gets nearly all of them wrong.

A batched version of the attack was then run on all 10 000 test images. The epsilon sweep in Figure **??** reveals a steep non-linear drop: already at $\varepsilon = 0.05$ accuracy falls to 23%, and above $\varepsilon = 0.15$ the model is essentially guessing randomly. A tiny, invisible change to the pixels — and the model is broken.

Task 2: Real-World Adversarial Examples

Idea 1 (Conservative): Physical Stickers on Traffic Signs

The starting point for this idea is a straightforward observation: if a neural network can be fooled by carefully crafted pixel perturbations in the digital domain, can the same principle be applied physically — by printing those perturbations on paper and placing them on a real-world object?

The threat scenario is concrete. An autonomous vehicle relies on a camera and a classification model to recognise traffic signs in real time. An attacker places algorithmically

designed stickers on a stop sign. To a human driver the sign still looks like a stop sign — the stickers resemble ordinary graffiti or vandalism. To the vehicle's model, however, the sign is now classified as something else entirely, for example a Speed Limit 45 sign. The vehicle does not stop.

This is not a hypothetical. In 2017, Eykholt et al. from the University of Washington, UC Berkeley and the University of Michigan demonstrated exactly this attack in a real-world experiment [1]. They computed adversarial perturbations using an algorithm called **Robust Physical Perturbations (RP2)**, printed the result as black-and-white stickers, and affixed them to a real stop sign on a real street. The attack was evaluated both in controlled lab conditions and in live drive-by tests with a moving vehicle approaching the sign from up to 75 metres away at speeds up to 20 mph, and at viewing angles between 0 and 60 degrees.

The results were striking. Under lab conditions the targeted misclassification rate was **100%**. In the real drive-by field test it reached **84.8%** of sampled video frames. The stop sign was consistently classified as Speed Limit 45 (LISA-CNN model) or Speed Limit 80 (GTSRB-CNN model). The attack remained robust across varying distances, angles and lighting conditions.

An important question arises: why does this work if the sign's distinctive octagonal shape is preserved? The answer reflects a fundamental difference between human and machine perception. A human recognises a stop sign primarily through its shape and the word "STOP". A neural network, however, operates on raw pixel values and has learned to associate specific colour and texture patterns with class labels. By perturbing those patterns in precisely the right way, the model's decision boundary is crossed even though the overall shape remains intact. The effectiveness of the attack therefore depends on how the target model was trained — models that rely heavily on texture and colour features are more vulnerable than those explicitly trained on shape representations.

To produce such attacks consistently and efficiently, the workflow would be: (1) obtain or train a model representing the target system, (2) compute the RP2 perturbation mask digitally, (3) print the mask on standard paper using a colour printer, accounting for printer colour reproduction errors via a Non-Printability Score constraint, and (4) cut and apply the stickers. The same printed perturbation can be reproduced at scale and applied to multiple signs, making the attack practical even with limited resources.

This example highlights a critical security implication: a printed piece of paper costing cents can reliably fool a deep learning system operating a vehicle at highway speeds. It motivates the broader research direction of *adversarial robustness* — designing models that remain reliable even when inputs have been deliberately manipulated.

Defence: Adversarial Training

The most direct and logical defence against physical adversarial attacks follows from understanding why they work in the first place. Standard models such as LISA-CNN and GTSRB-CNN were trained exclusively on clean datasets containing undisturbed traffic signs. Because the model never encountered deliberately crafted pixel perturbations during training, it has no reason to treat them as irrelevant noise — instead it incorporates them into its decision process as meaningful features.

The solution is to show the model adversarial examples during training itself. This approach is called **adversarial training**: the training set is augmented with perturbed versions of the original images, generated using the same attack algorithm (e.g. RP2 or FGSM). The model then learns to classify correctly despite the presence of such perturbations, because it has seen and survived them before.

Since the RP2 attack was specifically designed to remain effective across varying physical conditions — different viewing angles (0–60°), distances (up to 75 m) and lighting — a robust defence must match this scope. This means applying extended **data augmentation** during training: exposing the model to signs photographed from multiple angles, distances and under different lighting conditions, combined with adversarial perturbations. The model is thus forced to focus on the fundamental geometric features of the sign rather than surface-level texture and colour patterns that an attacker can manipulate.

Crucially, the same algorithm used to generate the attack can be repurposed as a defence tool. If the RP2 perturbation function is known, it can be run during training to continuously produce new adversarial examples on the fly — an approach known as **dynamic adversarial training**. This creates a feedback loop: the stronger the generated attack, the more robust the resulting model. As noted by Eykholt et al., understanding how physical attacks are constructed is itself "an important step towards developing resilient training algorithms" [1].

Idea 2 (Creative): Adversarial Clothing Against Surveillance Systems

The Scenario

A person walks through a public space monitored by surveillance cameras running an automated pedestrian detection system — a common setup in airports, shopping centres and city streets. The system is designed to detect and track every person in the frame. The goal of the attack is for the person to pass through undetected, without physically avoiding the cameras or covering their face.

The proposed approach: wear a jacket or t-shirt with a specially designed printed pattern that causes the detection model to fail to classify the wearer as a person.

Why a Simple Print Can Work

An important question arises immediately: if person detection models are trained to recognise the *shape* and *silhouette* of a human body, how can a surface print on clothing fool them? The silhouette is not changed by a print.

The answer lies in how deep neural networks process images internally. A model such as YOLO does not simply match a silhouette template. Instead, it builds a hierarchical representation across multiple layers: early layers detect edges and low-level textures, intermediate layers combine these into body-part-like features (torso, limbs), and final layers aggregate these features into a "person" classification. A carefully designed adversarial pattern does not need to remove the silhouette — it needs to corrupt the *intermediate feature maps* that the model relies on. By activating the wrong neurons in the middle layers, the pattern causes the final layer to output a low confidence score for the "person"

class, effectively making the wearer invisible to the detector while remaining fully visible to a human observer.

This was demonstrated in practice by Xu et al. in their work on adversarial T-shirt patches [2], where printed patterns caused person detectors to fail to detect the wearer in both lab conditions and real walk-by scenarios. Unlike rigid adversarial patches placed on flat surfaces, clothing deforms as the person moves — making the problem significantly harder and requiring the perturbation to be robust to fabric wrinkles and changes in viewing angle.

How to Produce This Consistently and Efficiently

The workflow for generating and testing adversarial clothing would be:

1. **Select a target model** — choose the person detection architecture to attack (e.g. YOLOv5, Faster R-CNN).
2. **Simulate clothing deformation** — use a differentiable renderer or a thin-plate spline transformation to simulate how the print deforms on fabric during movement. This makes the optimised pattern robust to wrinkles and pose changes.
3. **Optimise the adversarial pattern** — run a gradient-based optimisation to find a print that minimises the model's confidence score for the "person" class across many simulated viewpoints and deformation states.
4. **Print and test** — print the pattern on fabric using a standard textile printer and evaluate in real walk-by tests in front of the target camera setup.

Defence: Training on Adversarial Clothing

The same logic from Idea 1 applies here. A model that has never seen adversarial clothing patterns during training will be fooled by them. The defence is to augment the training data with images of people wearing adversarially generated patterns, forcing the model to learn features that are not disrupted by surface-level texture manipulation.

Additionally, **multi-cue detection** offers a more robust architectural defence: rather than relying solely on appearance features, the system can combine silhouette shape, motion patterns (from video), and depth information (from stereo cameras or LiDAR). An adversarial print can corrupt appearance features but cannot simultaneously fool all of these cues at once, making the overall system significantly harder to attack with a single wearable perturbation.

References

[1] Kevin Eykholt, Ivan Evtimov, Earlence Fernandes, Bo Li, Amir Raghunathan, Chuan Gao, Atul Prakash, Tadayoshi Kohno. *Robust Physical-World Attacks on Deep Learning Visual Classification.* arXiv:1707.08945, 2017. `https://arxiv.org/abs/1707.08945`

[2] Kaidi Xu, Gaoyuan Zhang, Sijia Liu, Quanfu Fan, Mengshu Sun, Hongge Chen, Pin-Yu Chen, Yanzhi Wang, Xue Lin. *Adversarial T-shirt! Evading Person Detectors in a Physical World.* arXiv:1910.11099, 2019. `https://arxiv.org/abs/1910.11099`